Pneumonia: Approach to a Case of Cough

Dr. Essam Kotb, Internal Medicine | Source: `11-Pneumonia Updated.pdf` (Google Drive, W4)

> ⚠️ **Text-only build (trial).** Made directly from Google Drive, which exposes only the slide **text**. The lecture's **images were not available**: CXR/CT pictures, the inpatient-vs-outpatient table, the CAP and HAP management tables, the classification mind-map graphic and the summary slide. Content that sat only in those images is rebuilt from standard guidance (IDSA/ATS 2019) and marked *[added]*. Upload the PDF and this note can be rebuilt with the original slides.
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> **Legend:** ⭐ = high-yield · *[added]* = not in the slide text (standard source, or a reconstruction of an image-only slide).
>
> 🔗 **Bridge from earlier respiratory lectures (same lecturer):**
>
> - **COPD/asthma:** pneumonia is a trigger of exacerbations.
> - **Lung abscess / bronchiectasis / CF:** aspiration and suppurative pneumonia.
> - **Pleura: parapneumonic effusion → empyema** (Light's criteria; pH < 7.2 → drain).
> - **Pulmonary vascular:** PE mimics pneumonia (pleuritic pain, fever, infiltrate).
> - **Chest investigations:** consolidation signs and air bronchograms.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Define pneumonia; alveolar and interstitial inflammation	Definition
Knowledge	⭐ Classify by **acquisition, microbe, site**	CAP vs HAP; lobar vs broncho vs interstitial
Knowledge	⭐ CAP organisms: **typical vs atypical**; HAP organisms	"Fisherman with lobar consolidation → ?"
Knowledge	⭐ Typical vs atypical: onset, symptoms, Gram stain, imaging, **penicillin response**	Pulse–temperature dissociation
Knowledge	Risk factors for pneumonia and aspiration	
Knowledge	Complications; prevention (pneumococcal and influenza vaccine, smoking cessation)	Vaccinate before discharge
Skill	Interpret findings; read CXR/CT (**lobar consolidation, air bronchogram, reticulonodular infiltrates**)	Image MCQ
Skill	⭐ **CURB-65** → outpatient / ward / ICU	Lecture case 2
Skill	⭐ Empiric regimen by risk and severity; **MDR risk → dual/triple IV therapy** for HAP	Regimen MCQ

Main subtopics: CAP · HAP · atypical pneumonia · aspiration pneumonia (definition, types, pathogenesis, presentation, diagnosis, management).

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2. Main Content (Lecture Order)

2.1 Definition & Transmission

Pneumonia = respiratory infection with **inflammation of the alveolar space and/or the interstitial tissue** of the lungs.

- Most commonly from **aspiration of airborne pathogens** (mainly **bacteria**; also viruses and fungi).
- Can also result from **aspiration of stomach contents**.

2.2 Classification ⭐

Basis	Types
Acquisition	**CAP** vs **HAP** (± healthcare-associated; pneumonia in the **immunocompromised host**)
Microbe	Bacterial vs viral vs fungal
Site	**Lobar** vs **bronchopneumonia** vs **interstitial (atypical)**
Aetiology (mind-map)	**Primary** vs **secondary** vs **suppurative**
Special	**Aspiration**

[added] Lobar = a whole lobe, uniform (classically pneumococcus). Bronchopneumonia = patchy, centred on bronchioles, often bilateral and basal (S. aureus, H. influenzae, Gram-negatives, elderly/bedbound). Interstitial = alveolar walls/septa (viruses, Mycoplasma).

2.3 Typical CAP: Organisms ⭐

- ⭐ ***Streptococcus pneumoniae*: 60%** (the commonest cause of CAP)
- ***Haemophilus influenzae*: 15%**
- **Aerobic Gram-negative rods: 6–10%**: *Klebsiella* and other Enterobacteriaceae
- ***Staphylococcus aureus*: 2–10%**

2.4 Risk Factors

- **Old age, immobility**
- **Chronic disease:** - pre-existing **cardiopulmonary** disease (asthma, **COPD**, **HF**) - **airway abnormalities** (bronchiectasis, space-occupying lesions, **CF**)
- **Immunosuppression**: **HIV**, **diabetes**, immunosuppressive drugs
- ⭐ **Impaired airway protection** (aspiration risk): **altered consciousness** from stroke, seizure, anaesthesia, drugs, **alcohol**
- **Environment**: **crowding**; **contaminated water systems** (hotels, cruise ships → *Legionella*)
- **Surgery**

2.5 Typical CAP: Clinical Picture ⭐

- **Symptoms:** - **acute** fever with **shaking chills (rigors)** - **productive cough** with thick **purulent** sputum (*[added]* "rusty" in pneumococcus; "currant jelly" in *Klebsiella*) - **pleuritic chest pain** - dyspnoea
- **Signs:** - tachycardia, tachypnoea - ⭐ **consolidation signs**: **late inspiratory crackles, bronchial breath sounds, ↑ tactile and vocal fremitus, dullness to percussion** - **pleural friction rub** (pleurisy)

- **CXR: ⭐ lobar consolidation**. **Multilobar consolidation = very serious illness**.

> *Slides "Lobar PNA" ×2: CXR images not available.* **[image missing]**

2.6 Atypical CAP: "Walking Pneumonia" ⭐

- **Organisms:** - ⭐ ***Mycoplasma pneumoniae* (most common)** - *Chlamydia (Chlamydophila) pneumoniae* - ***Chlamydia psittaci*** (*[added]* birds) - ***Coxiella burnetii*** (Q fever; *[added]* farm animals) - ***Legionella*** spp. - **viruses**: influenza A/B, adenovirus, parainfluenza, RSV
- **Symptoms:** - **insidious** onset: headache, sore throat, fatigue, myalgia - ⭐ **dry cough** (no sputum) - fever, but **chills are uncommon**
- **Signs:** - ⭐ **pulse–temperature dissociation**: a **normal pulse despite high fever** suggests atypical CAP - wheeze, rhonchi, crackles
- **CXR: ⭐ diffuse reticulonodular infiltrates**; **absent or minimal consolidation**.

> *Two "Atypical CAP" image slides not available.* **[image missing]**

Typical vs atypical CAP at a glance

TYPICAL (lobar)

S. pneumoniae 60% · H. influenzae 15%
Klebsiella / GNR 6–10% · S. aureus 2–10%

Acute fever + SHAKING CHILLS
PRODUCTIVE purulent sputum
Pleuritic chest pain, dyspnoea

Consolidation signs:
bronchial breathing, ↑ fremitus, dullness,
late inspiratory crackles ± pleural rub

CXR: LOBAR consolidation, air bronchogram
(multilobar = severe)

Gram stain: organisms + neutrophils
Responds to β-lactams (penicillin)

ATYPICAL ("walking pneumonia")

Mycoplasma (most common) · Chlamydia
pneumoniae / psittaci · Coxiella · Legionella · viruses

INSIDIOUS: headache, sore throat, myalgia
DRY cough · fever, chills uncommon
Pulse–temperature dissociation

Few chest signs:
wheeze, rhonchi, crackles
(exam milder than the CXR)

CXR: diffuse RETICULONODULAR infiltrate
little or no consolidation

Gram stain: few organisms
NO response to penicillin → macrolide/doxy/FQ

Added diagram: typical vs atypical CAP

[added] Organism clues:

Clue	Think
Young adult, dormitory, **cold agglutinins**, bullous myringitis, erythema multiforme	*Mycoplasma*
Hyponatraemia, diarrhoea, confusion, ↑ LFTs, hotel/cruise/air-conditioning	*Legionella* (**urinary antigen**)
Birds (parrots)	*C. psittaci*
Farm animals, parturient sheep/goats	*Coxiella* (Q fever)
Alcoholic, **currant-jelly sputum**, upper-lobe bulging fissure	*Klebsiella*
After influenza, cavitation, pneumatoceles	*S. aureus*
COPD smoker	*H. influenzae*, *Moraxella*
Aspiration, foul sputum, right lower lobe	Anaerobes (*Fusobacterium*, *Bacteroides*)

2.7 Diagnostics ⭐

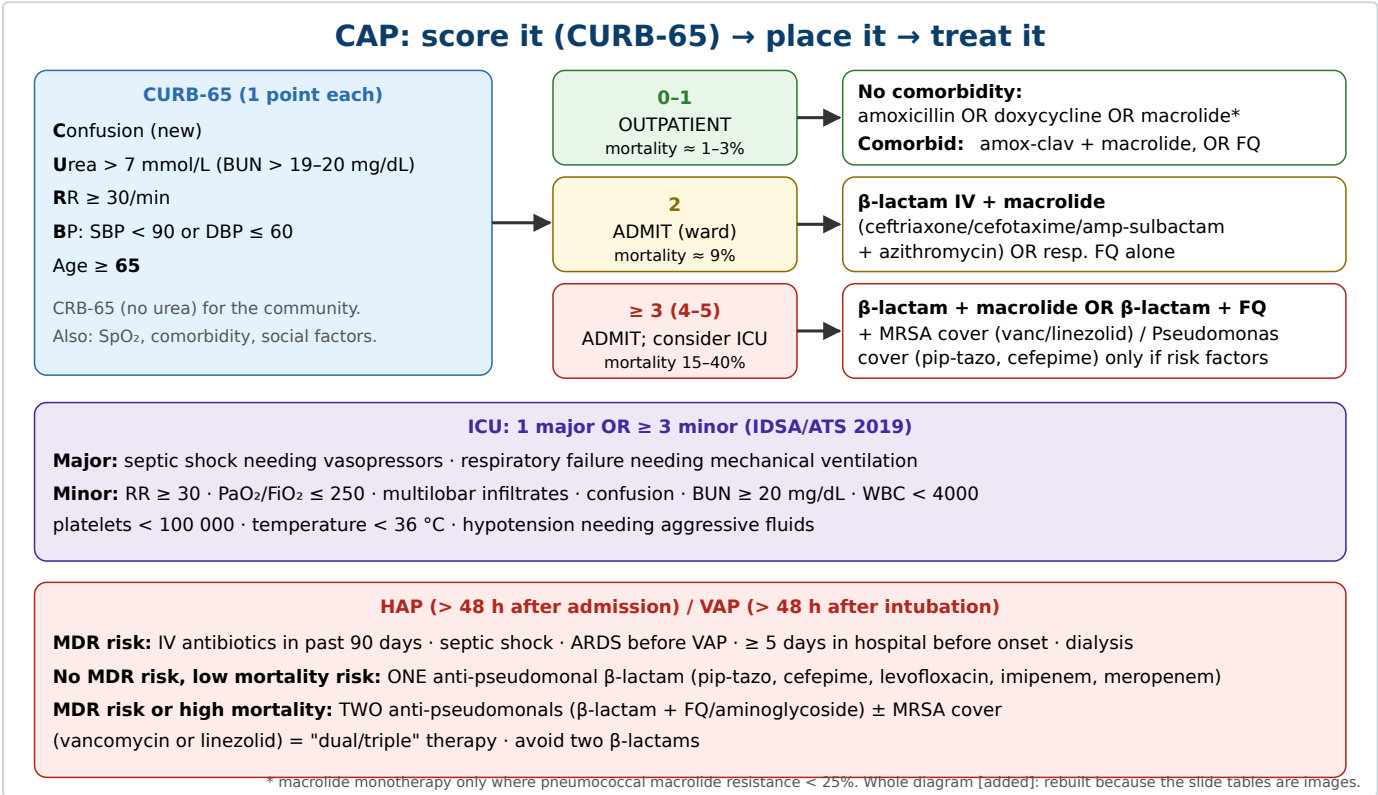
- **CXR (PA + lateral)**: first-line imaging
- **CBC**; **↑ CRP, ↑ ESR**; ⭐ **↑ serum procalcitonin** (bacterial)
- **Chest CT** (without contrast)
- **Sputum Gram stain and culture**; **blood cultures**
- **Influenza (flu swab) and COVID-19 testing**
- **↓ SpO_2** (pulse oximetry)
- ⭐ **Urinary antigen for *Legionella*** (when suspected) (*[added]* also pneumococcal urinary antigen)

> *Slides "CT Chest" and "Air Bronchogram": images not available.* **[image missing]**

[added] Air bronchogram = air-filled bronchi visible against opacified (fluid-filled) alveoli. It is the hallmark of alveolar consolidation and helps distinguish it from effusion or collapse.

2.8 Management: Where to Treat ⭐

The slide text says: **"Hospitalisation is indicated" / "Consider ICU admission"**. The scoring table itself was an **image** and was not available. Rebuilt with **CURB-65** (named in the objectives) and the IDSA/ATS ICU criteria *[added]*:



Added diagram: CURB-65, disposition, empiric regimens, HAP/VAP

CURB-65 *[added detail]*	Disposition
0–1	**Outpatient**
2	**Admit** (short stay or ward)
≥ 3	**Admit; consider ICU** (especially 4–5)

2.9 Empiric Management ⭐

General (slide text):

- **Empiric antibiotics by severity and risk factors** (e.g., VAP vs CAP)
- **Antivirals** for influenza and COVID-19 as indicated
- **Identify and manage sepsis**; **fluid resuscitation** for septic shock
- **Respiratory support** as needed; **supplemental O_2** for hypoxia

Outpatient options visible in the text: "Amoxicillin" OR **"combination (amoxicillin + macrolide)"**. The rest of the CAP table was an image and is rebuilt below.

Setting	Regimen *[added: IDSA/ATS 2019, except the amoxicillin lines]*
Outpatient, healthy	⭐ **Amoxicillin** (slide) or doxycycline; macrolide only if local resistance < 25%
Outpatient, comorbid (COPD, DM, HF, CKD, alcohol, malignancy)	**Amoxicillin-clavulanate (or cephalosporin) + macrolide** (slide: "amoxicillin + macrolide"), OR **respiratory FQ** (levofloxacin, moxifloxacin)
Inpatient, non-severe	⭐ **IV β-lactam (ceftriaxone / cefotaxime / ampicillin-sulbactam) + macrolide (azithromycin)**, OR respiratory FQ alone
Severe / ICU	**β-lactam + macrolide** or **β-lactam + FQ**; add **MRSA** (vancomycin/linezolid) or **Pseudomonas** (pip-tazo, cefepime, meropenem) cover only with risk factors

2.10 Prevention

- ⭐ **Pneumococcal vaccine: give to ALL patients with CAP before discharge** unless already vaccinated.
- **Influenza vaccine.**
- **Smoking cessation.**

2.11 Hospital-Acquired Pneumonia ⭐

- **HAP:** onset **> 48 h after admission**, with **no evidence of the disease at admission**.
- *[added]* **VAP:** > 48 h after endotracheal intubation.
- The HAP management table was an image. Rebuilt from the objective "*assess MDR risk factors to select IV dual/triple empiric regimens*":

	Regimen *[added]*
MDR risk factors	IV antibiotics in the **past 90 days**, **septic shock**, ARDS before VAP, **≥ 5 days in hospital** before onset, dialysis
No MDR risk, low mortality	**One** anti-pseudomonal agent: **pip-tazo, cefepime, levofloxacin, imipenem or meropenem**
MDR risk / high mortality	⭐ **Two anti-pseudomonal agents from different classes** (β-lactam + FQ or aminoglycoside) **± MRSA cover (vancomycin or linezolid)** = "dual/triple"

[added] HAP organisms: Pseudomonas, MRSA, Klebsiella, E. coli, Enterobacter, Acinetobacter.

2.12 Aspiration Pneumonia *(listed as a subtopic; no detail in the slide text)*

[added]:

- **Risk:** reduced consciousness (stroke, seizure, alcohol, sedation, anaesthesia), dysphagia, poor dentition.
- **Site: right lower lobe** when upright (superior segment of the lower lobe / posterior upper lobe when supine).
- **Aspiration pneumonitis** (chemical, Mendelson) vs **aspiration pneumonia** (anaerobes + oral flora).
- **Treatment:** ampicillin-sulbactam or clindamycin; risk of **lung abscess / empyema**.

2.13 Complications

- **Parapneumonic pleuritis**
- **Parapneumonic effusion** → ⭐ **empyema**
- **Lung abscess**
- **ARDS** → **respiratory failure**
- **Sepsis**

> *Summary slide: image, not available.* **[image missing]**

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3. High-Yield Comparison Tables

Table A: Typical vs Atypical CAP

	Typical	Atypical
Organisms	**S. pneumoniae 60%**, H. influenzae 15%, GNR 6–10%, S. aureus 2–10%	**Mycoplasma (commonest)**, Chlamydia, Coxiella, Legionella, viruses
Onset	**Acute**	**Insidious**
Fever / chills	High fever + **shaking chills**	Fever, **chills uncommon**
Cough	**Productive, purulent**	**Dry**
Pulse	Tachycardia	**Pulse–temperature dissociation**
Exam	**Consolidation signs**	Wheeze, rhonchi, crackles (mild)
CXR	**Lobar consolidation**	**Diffuse reticulonodular**, minimal consolidation
Gram stain	Organisms + PMNs *[added]*	Few organisms *[added]*
Penicillin	**Responds**	**No response** (no cell wall / intracellular) *[added]*

Table B: CAP vs HAP

	CAP	HAP / VAP
Definition	Acquired in the community	**> 48 h after admission** (VAP: > 48 h after intubation)
Organisms	Pneumococcus, H. influenzae, atypicals	**Pseudomonas, MRSA**, GNR, Acinetobacter *[added]*
Empiric	Amoxicillin ± macrolide → β-lactam + macrolide	**Anti-pseudomonal**: 1 agent, or **2 + MRSA cover** if MDR risk *[added]*

Table C: Consolidation vs Effusion on Exam *[added; links to Pleura lecture]*

Sign	Consolidation	Effusion
Percussion	**Dull**	**Stony dull**
Fremitus	**↑**	**↓**
Breath sounds	**Bronchial**	**↓ / absent**
Crackles	Present	Absent

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Pneumonia → bacteria	Viruses (influenza, COVID-19) are common: test and give antivirals
Fever → tachycardia	**Pulse–temperature dissociation** in atypicals (*[added]* Legionella, Mycoplasma)
Penicillin for CAP	**Atypicals don't respond**: add a macrolide/doxycycline or use an FQ
Big CXR = sick patient	Atypicals: **CXR often looks worse than the exam** *[added]*
One lobe = typical	**Multilobar = severe**
Vaccinate later	⭐ **Pneumococcal vaccine before discharge**
HAP = any hospital pneumonia	Only if onset is **> 48 h after admission** and absent at admission
Single agent for HAP	**MDR risk → dual anti-pseudomonal ± MRSA**

5. Clinical Correlations

- **Elderly diabetic smoker with rigors, rusty sputum and right upper lobe consolidation** → pneumococcal CAP → CURB-65 → ceftriaxone + azithromycin if admitted.
- **Student with weeks of dry cough, headache, sore throat, normal pulse, patchy bilateral infiltrates** → *Mycoplasma* → macrolide or doxycycline.
- **Hotel guest with pneumonia, diarrhoea, confusion, Na 128** → *Legionella* → **urinary antigen**; FQ or azithromycin.
- **Post-stroke patient with right lower lobe pneumonia and foul sputum** → aspiration → amp-sulbactam.
- **Day 6 of admission, new fever and infiltrate after IV antibiotics** → **HAP with MDR risk** → pip-tazo + FQ ± vancomycin.
- **Persisting fever on antibiotics with an enlarging effusion** → tap for **empyema** (pH < 7.2 → drain) *[Pleura lecture link]*.

6. Rapid Revision (Last-Day)

1. Pneumonia = alveolar and/or interstitial inflammation, mostly from aspirated airborne pathogens.
2. Classify by **acquisition** (CAP/HAP), **microbe**, **site** (lobar/broncho/interstitial), **aspiration**.
3. Typical CAP: **S. pneumoniae 60%**, H. influenzae 15%, GNR 6–10%, S. aureus 2–10%.
4. Risks: age, immobility, COPD/asthma/HF, bronchiectasis/CF, HIV/DM/immunosuppression, **↓ consciousness** (aspiration), crowding, **contaminated water**, surgery.

5. Typical: acute fever, **rigors**, purulent sputum, pleuritic pain; **dull, ↑ fremitus, bronchial breathing, crackles**; **lobar consolidation** (multilobar = severe).
6. Atypical: **Mycoplasma** commonest; insidious, **dry cough**, few chills, **pulse–temperature dissociation**; **reticulonodular** CXR.
7. Tests: CXR, CBC, CRP/ESR, **procalcitonin**, sputum and blood cultures, flu/COVID swab, SpO_2, **Legionella urinary antigen**, CT.
8. **CURB-65**: Confusion, Urea > 7, RR ≥ 30, BP < 90/≤ 60, age ≥ 65 → **0–1 home, 2 admit, ≥ 3 ICU consideration** *[added detail]*.
9. Outpatient: **amoxicillin**, or **amoxicillin + macrolide** if comorbid (slide). Inpatient: **β-lactam + macrolide** or resp. FQ *[added]*.
10. Also: antivirals as indicated, sepsis care, fluids, O_2, respiratory support.
11. **Pneumococcal vaccine before discharge**; influenza vaccine; stop smoking.
12. **HAP > 48 h** after admission. **MDR risk → dual anti-pseudomonal ± MRSA** *[added detail]*.
13. Complications: pleuritis, **parapneumonic effusion/empyema**, abscess, **ARDS**, respiratory failure, **sepsis**.

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7. Exam-Style Questions

Q1 (lecture case 1). A 53-year-old commercial fisherman has had fever, chills, malaise and **yellow-green productive cough for 1 day**. T 38.5, HR 96, RR 26, BP 98/62. **↑ fremitus and bronchial breath sounds**; CXR shows **right upper lobe consolidation**. He takes metoprolol and warfarin. Most likely pathogen? a. M. tuberculosis · b. Pseudomonas · c. Fusobacterium · d. Mycoplasma · e. Legionella · f. S. aureus · g. S. pyogenes · h. H. influenzae · i. Histoplasma · **j. S. pneumoniae**

> **Answer: j. S. pneumoniae.** *[reasoning added; key not in slide text]*
>
> - **Acute onset (1 day)**, rigors, purulent sputum and **lobar consolidation** = typical CAP. Pneumococcus is the commonest cause (60%).
> - The occupation is a distractor (Blastomyces/Histoplasma suggest a subacute course).
> - The **metoprolol** explains the relatively low pulse.
> - **Legionella** would bring GI symptoms, hyponatraemia and an exposure history.
> - **Mycoplasma** is insidious with a dry cough.
> - **TB** is chronic.

Q2 (lecture case 2). A 67-year-old woman (HTN, DM, ex-smoker) has 2 days of fever, chest pain and purulent cough. T 39, HR 110, **RR 33**, **BP 143/88**, SpO_2 94%, crackles over the right upper zone, WBC 12 300, **BUN 15 mg/dL**, alert. Next step? a. Outpatient moxifloxacin · b. Inpatient cefepime + azithromycin + gentamicin · c. ICU, ceftriaxone + azithromycin · d. Outpatient azithromycin + amox-clav · **e. Inpatient azithromycin + cefotaxime** · f. ICU, amp-sulbactam + levofloxacin · g. Outpatient azithromycin · h. Inpatient ceftriaxone alone · i. ICU, aztreonam + levofloxacin

> **Answer: e.** *[reasoning added]*
>
> - **CURB-65 = 2**: age ≥ 65 (1) + RR ≥ 30 (1). She is not confused, BUN 15 is < 20, and BP is normal. Score 2 → **admit to the ward**.
> - Only **one** IDSA minor criterion (RR ≥ 30) and no major criterion → **no ICU**.
> - Non-severe inpatient CAP → **β-lactam (cefotaxime/ceftriaxone) + macrolide (azithromycin)**.
> - b is an MDR/HAP-type regimen.
> - h (ceftriaxone alone) misses atypical cover.

Q3. A patient has a high fever but a pulse of 78, a dry cough for 10 days, headache and diffuse reticulonodular infiltrates. What is the best empiric agent? A. Amoxicillin · B. **Azithromycin (or doxycycline)** · C. Ceftriaxone · D. Vancomycin · E. Metronidazole

> **Answer: B.**
> - Atypical CAP (**pulse–temperature dissociation**). β-lactams don't cover atypicals.

Q4. Seven days into a hospital stay, after IV ceftriaxone for cellulitis, a patient develops fever, purulent sputum and a new infiltrate. What is the empiric therapy? A. Amoxicillin · B. Ceftriaxone + azithromycin · C. **Piperacillin-tazobactam + levofloxacin + vancomycin** · D. Oral doxycycline · E. Clindamycin alone

> **Answer: C.**
> - **HAP with MDR risk** (IV antibiotics in the past 90 days, ≥ 5 days in hospital) → **dual anti-pseudomonal + MRSA cover** *[added: IDSA 2016]*.

Q5. Which finding indicates **severe** pneumonia on CXR? A. Air bronchogram · B. Reticulonodular pattern · C. **Multilobar consolidation** · D. Blunted costophrenic angle · E. Hyperinflation

> **Answer: C** (slide: "multilobar consolidation indicates very serious illness").

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8. Exam Pitfalls & Traps

- **Pneumococcus is still the commonest cause of CAP**, typical and overall. Mycoplasma is only the commonest **atypical** cause.
- **Pulse–temperature dissociation** points to atypicals. Watch for β-blockers (as in case 1) faking it.
- **Ceftriaxone alone** for admitted CAP misses atypicals; add azithromycin (or use an FQ).
- **HAP timing:** onset **> 48 h** after admission **and** absent at admission.
- **Pneumococcal vaccine before discharge** is a slide-specific management point.
- **Legionella:** diagnose with a **urinary antigen**. It doesn't Gram stain well.
- **This trial note is incomplete by design:** the CURB-65 thresholds, outpatient/inpatient/ICU table, CAP/HAP regimen tables and every X-ray were slide images. What appears here is the standard guideline content (*[added]*), **not** necessarily the lecturer's exact table. Check against the original slides before the exam.